Coordination of Spiral Waves in the Motor Cortex

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A thesis presented for the degree of
Bachelor of Science and Bachelor of Advanced Studies (Honours)

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May 17, 2024

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Abstract

Advances in brain-imaging techniques have revealed highly organized spatiotemporal activity patterns as core features of the processes central to our understanding of cognition. Although the presence of spiral-shaped brain waves in functional magnetic resonance imaging (fMRI) recordings is well-documented, their role in cognitive processing and the organizational principles underlying their dynamics are still uncertain. Here, based on human fMRI recordings, we investigate the mechanistic role of spiral waves in coordinating turbulent-like cortical dynamics and their functional correlates with motor behaviours. Consistent with previous studies, we find the properties of these spiral waves, such as their rotational directions and locations, to be task-relevant and valuable in the neural decoding of motor tasks. By observing the flow of brain activity, we elucidate hubs of activity surrounded by pairs of spiral waves with the description of elementary rules that mechanistically describe the functional role of these spirals. Additionally, using a mathematical model, we demonstrate that unique interactions among an array of spiral waves distributed across the cortex may successfully capture the structure of the underlying brain function. Specifically, we can reliably predict the complex dynamics of the human cortex solely based on the spatial configurations of the spiral wave array, which consists of only a few dozen spiral wave patterns. These findings provide novel evidence supporting the mechanistic and functional importance of cortical spiral waves underlying motor function in the human brain.

Statement of Originality

I, Adam Karvon, certify that this thesis contains work carried out by myself except where otherwise acknowledged. This thesis is submitted in fulfilment of the requirements for the conferral of the degree Bachelor of Science and Bachelor of Advanced Studies, from the University of Sydney.

Signature: Date: 17/05/2024

Acknowledgements

I would like to thank my supervisor, Dr Pulin Gong, for giving me the opportunity to diligently and creatively contribute to this research project and his continued guidance throughout the year. I am grateful to Ben Yiben Xu, for his reassurance, guidance and deep knowledge on the subject. I also appreciate the support I have received this year from members of the complex systems group specifically Kevin Qu, Andrew Ly and Yifan Chen who formed a great community to be a part of. Finally, I would like to thank my family and my loving girlfriend for their unconditional support in all my life, career and academic pursuits.

Statement of Student Contributions

The theoretical foundation behind interactions of spiral waves in the brain was developed by the group led by my supervisor, A/Prof. Pulin Gong. Data pre-processing through the Brain-Vortex toolbox was designed by the group for a study on spiral waves and their relation to cognitive tasks.

I made significant modifications to the Brain-Vortex toolbox to accommodate spiral wave detection in motor task data. Additionally, I conducted analyses of how spiral waves respond to the onset of tasks using a density-based clustering algorithm. Inspired by a paper recommended by my supervisor, I independently integrated particle tracking software into MATLAB to monitor the movement and characteristics of spiral wave centres (Chapter 2).

I adapted the methodology for generating surrogate data and similarity filtering of spiral waves from prior research and independently wrote the code for generating spiral-centre-density maps and a linear classifier. I further devised a new method to analyse the magnitudes and directions of underlying wave vectors of brain activity, implementing these processes in MATLAB. (Chapter 3).

Guided by my supervisor, I formulated a new mathematical model to describe the interactions of phase singularities. I independently devised techniques to pinpoint the rotational phase of spiral waves and refined the model through gradient descent, focusing on the radial influence of phase singularities (Chapter 4).

All figures were generated by code I wrote, unless indicated otherwise.

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Introduction

Before an idea is a thought, it is an interaction of neurons in the brain. Just as dynamic and chaotic as the ideas they create, human brains demonstrate a level of complexity imitated only by the most intricate excitable systems, such as chemical reactions, the formation of galaxies, or the turbulent flow of fluids. Recent advancements in macroscale recording modalities, such as functional magnetic resonance imaging (fMRI), have only accentuated brain activity's complex spatiotemporal nature. A critical feature in excitable systems is the presence of spatiotemporal patterns such as sources, sinks, and spiral waves. These structured patterns, particularly spiral waves, have been shown to capture much of the structure of the systems in which they appear. Given the occurrence of spiral waves in many neural contexts, one might ask what makes spiral waves in neural recordings any different from their appearance in other chaotic systems. This thesis explores exactly this intriguing aspect of neural coordination, elucidating the specific behaviours, functional significance, and interactions of spiral waves that remain enigmatic.

This research is poised at the cutting edge of neuroscience, applying advanced imaging and computational techniques to uncover the mysteries of cortical activity patterns. The methodologies employed—from detailed fMRI data analysis to sophisticated mathematical modelling of neural dynamics—represent a significant innovation in studying brain function. Moreover, the interdisciplinary approach, drawing on insights from optical physics, underscores the complex and multifaceted nature of brain dynamics research.

By studying brain activity during motor tasks, this thesis aims to dissect the properties and functional implications of spiral waves in the motor cortex. It hypothesizes that these waves are integral to the coordination of the neural activity behind motor functions, proposing that a deeper understanding of their dynamics could lead to novel insights into brain function and dysfunction. Through a combined approach of experimental data analysis and mathematical modelling, this thesis endeavours to construct a comprehensive framework that elucidates the role of spiral waves in motor cortex dynamics

Here, we set forth several objectives for thoroughly investigating this hypothesis. In Chapter 1, we will survey the existing literature on brain dynamics, building up from the smallest scales and exploring the notable interdisciplinary relevance of spiral waves to the exploration of chaotic systems. Chapter 2 will describe the methodology of mapping spiral waves during various motor tasks using high-resolution fMRI data, exploring both quantitatively and qualitatively the properties of spiral waves and their dynamics in response to motor tasks. Chapter 3 will explore the functional

relevance of spiral waves by decoding the information contained within their characteristic behaviours, demonstrating the mechanisms through which they facilitate distributed coordination of neural activity. Chapter 4 depicts the development of mathematical models that describe the interactions between multiple spiral waves to simulate and predict the dynamics observed in experimental recordings. Chapter 5 interprets the results within the framework of existing knowledge and explores the implications for future research.

The findings from this research are expected to advance our understanding of the fundamental mechanisms by which the brain coordinates complex motor actions. By clarifying the role of spiral waves in the motor cortex, this thesis contributes to the broader field of neuroscience, potentially impacting future research into brain function and dysfunction. Furthermore, the insights gained may inform the development of novel therapeutic strategies for neurological disorders involving motor function impairment. By providing a comprehensive overview of both the theoretical and practical aspects of spiral wave dynamics in the motor cortex, this introductory chapter sets the stage for a detailed exploration of one of the brain's most intriguing yet underexplored phenomena.

Chapter 1

Literature Review

This thesis will explore the large scale coordination of neural activity within the motor cortex. As brain activity complexity starts at the level of individual neurons, Section 1.1 begins by examining patterns across the various scales of brain activity imaging, establishing essential concepts for interpreting later results. Section 1.2 focuses on the different spatiotemporal phenomena central to macroscopic imaging methods, placing this research in the context of similar studies. Finally, Section 1.3 explains the crucial role that spiral waves play in various excitable systems where they are observed. The literature presented in this section lay the groundwork for our methodology and rationale for studying spiral waves in the brain.

1.1 Spatiotemporal patterns of neural activity across multiple scales

Brain activity is typically categorised into three main types according to its spatial dimension. Microscale activity involves individual neurons and occurs at less than 0.1 mm. Mesoscale activity encompasses small neural networks ranging from 0.1 mm to 10 mm. Finally, macroscale activity spans across larger brain regions, exceeding 10 mm in scale. Understanding the multi-scale nature of neural activity is pivotal in comprehending how the brain processes information dynamically and in a distributed manner. This section delves into the techniques employed to capture neural dynamics at the microscale, revealing intricate details of neuronal activity and cellular processes.[1]

1.1.1 Waves of spiking patterns in neurons

Microscale recordings provide insights into the neural activity at the level of individual neurons, offering a detailed understanding of the basic building blocks of brain function. Single-unit Electrode Recordings have captured direct measurements of voltage potentials across the membrane of individual neurons. These techniques have revealed that neurons in awake animals exhibit highly irregular firing patterns in the absence of external stimuli.[2] This spiking variability exceeds what would be expected from random, independent inputs, indicating a more complex underlying neural process.[1][3] Further studies modelling neurons and their synaptic inputs

have shown that the generation of local, propagating waves of activity, which can occur through balanced excitatory and inhibitory inputs, may provide a mechanism for explaining the irregular firing dynamics of in vivo neurons.[4] This mechanism is based on the observations of model neurons and neurons in vitro, which can imitate in vivo neurons through synchronised inputs that induce rapid changes and strong nonlinearities in the membrane potential.[3][5]

In the motor cortex, studies have utilised high density microelectrode arrays (MEAs) to capture detailed activity from many neurons simultaneously. MEAs enable the recording of local field potentials (LFPs) and multiunit activity (MUA), providing a comprehensive view of the electrophysiological landscape of the cortex. This has revealed the spiking activity of a subset of motor cortical neurons in primates that are spatially coordinated in a manner that closely matches wave propagation observed in the (15-40 Hz) oscillatory band of the LFP. This coordination was particularly evident during motor tasks, suggesting that these propagating waves play a significant role in motor control.[6] These waves travelled in specific directions with consistent speeds, revealing stage-specific patterns that reflect the organised structure of local synaptic inputs.[6]

Further studies using MEAs in humans have analysed neuronal activity in the motor cortex during epileptic seizures. Findings have suggested that distinct stages of seizure propagation in the motor cortex are characterised by specific spatiotemporal dynamics, indicating organised and consistent propagation patterns across different stages.[7] The study emphasises that individual spike-wave discharges and gamma oscillation cycles propagate with stage-dependent characteristics such as direction, speed, and directionality index. These characteristics are influenced by the discrete stages of seizure propagation, reflecting the complex and organised structure of local synaptic inputs. The propagation patterns are more complex than simple plane waves, with examples of collapsing or expanding waves. These patterns are consistent within patients but vary across different patients, likely due to differences in etiology, seizure onset zones, and electrode locations. The study highlights the need for further research to understand the physiological substrates underlying these propagation patterns.[7]

The use of high-density MEAs in these studies demonstrates the potential of advanced microscale recording techniques to uncover the fundamental mechanisms underlying cortical dynamics. By capturing the nuanced behaviours of individual neurons and their networks, these techniques offer profound insights into the cellular processes that underpin the broader spatiotemporal patterns observed in the brain.

1.1.2 Mesoscale dynamics in the motor cortex

The collective behaviour of neurons, as observed through mesoscale recording techniques such as voltage-sensitive dye (VSD), underscores the intricate spatiotemporal dynamics fundamental to cortical function and motor control. Studies have highlighted the capability of VSD imaging to capture subthreshold depolarizations and propagating waves across the neocortex, which establish a spatiotemporal framework for neuronal activity.[8] These waves enhance cortical excitability and synchronize neuronal firing, crucial for integrating sensory inputs with motor outputs. Findings have revealed that sensory responses in the motor cortex are modulated

by behavioural states, showing smaller and more localized responses during active whisking compared to quiet wakefulness in rodents.[9] This modulation indicates that the motor cortex dynamically integrates sensory information based on the current motor state, facilitating precise motor control.

Dimensionality reduction techniques used to study underlying structures in the brain have provided further insights into the spatiotemporal dynamics of neural populations during motor tasks. Using joint principal component analysis (jPCA), it was discovered that neural activity in the motor cortex exhibits rotational dynamics during movement initiation.[10] These rotations, representing low-dimensional trajectories in neural state space, suggest that the motor cortex operates as a dynamical system. This system organizes preparatory activity into rotational patterns that evolve predictably to produce coordinated motor outputs.[10] This finding challenges traditional representational models, which posit that motor cortex neurons encode specific movement parameters like direction and velocity. Instead, the dynamical systems approach emphasizes that the coordinated evolution of neural population activity is central to generating motor behaviour.

Building on this dynamical systems perspective, further studies have compared representational and dynamical models of motor cortex activity. One study used simulations and empirical data to show that while representational models can account for some aspects of neuronal tuning, they fall short of explaining the rotational structure observed in neural population activity.[11] By implementing a covariance-matched permutation test, it was demonstrated that the rotational dynamics observed in motor cortex data are more consistent with a dynamical model. This model posits that muscle activity is generated by a combination of oscillatory components, each with condition-specific phases and amplitudes, set during the preparatory period.[11] This supports the view that the motor cortex functions as a dynamical system that transforms preparatory neural states into coordinated motor actions.

Extensions of these findings are relevant to the study of neurological disorders, particularly Amyotrophic Lateral Sclerosis (ALS).[12] In individuals with ALS, this modelling showed neural population dynamics in the motor cortex during movement exhibited similar rotational patterns to those seen in healthy individuals. This suggests that the fundamental principles of motor cortex dynamics remain intact despite motor impairments.[12] These observations underscore the robustness of the dynamical system framework in explaining motor control. Collectively, these studies highlight the significance of mesoscale neuronal dynamics in bridging the gap between micro-level neuronal activity and macro-level motor behaviour, ultimately enhancing our understanding and ability to manipulate the neural basis of motor control.

1.1.3 Macroscale brain oscillations

Macroscale recordings provide a comprehensive view of brain activity, capturing the dynamics of entire brain regions during various cognitive states and tasks. Electroencephalography (EEG), Electrocorticography (ECoG), Magnetoencephalography (MEG), and functional MRI (fMRI) are indispensable for investigating the spatiotemporal dynamics of neural activity in the brain. These modalities offer complementary insights into how oscillatory activity across various frequencies con-

tributes to brain function.[13][14]

Neuronal oscillatory activity spans from slow to fast frequency bands, each associated with distinct cognitive functions and states, such as sleep and active cognition.[15][16][17] The interplay between these frequencies, including mechanisms like cross-frequency coupling, is critical for understanding neural communication and functional integration.[18] EEG provides a broad, non-invasive overview of the brain's electrical activity, ECoG offers more localised measurements due to its proximity to the cortex, and MEG detects the minute magnetic fields generated by neuronal currents, providing a unique perspective on the tangential components of this activity. In parallel, fMRI complements these techniques by mapping the functional architecture through metabolic changes, offering a different dimension of spatial resolution and functional connectivity. Together, these techniques furnish a multifaceted view of the brain's operational landscape, crucial for linking macroscale brain activity with behavioural outcomes and elucidating the underlying mechanisms of cognitive functions. They collectively highlight how large-scale oscillations arising from the coordinated firing of neural populations modulate the activity of individual neurons, thus advancing our comprehension of the complex spatiotemporal patterns that define brain function.[1]

Despite their widespread use, macroscale recording techniques face several challenges, such as distortions in EEG due to the skull and scalp and the need to enhance the signal-to-noise ratio in MEG for more precise brain activity mapping. Technology and data processing advances continue to address these issues, improving the accuracy and utility of these methods in clinical and research settings.[19] Macroscale recordings are crucial for linking brain activity to behavioural outputs and understanding the neural underpinnings of cognitive functions, offering a holistic view of the brain's operational landscape.

1.2 Cortex-wide spatiotemporal patterns

1.2.1 Propagating brain waves

Increasing evidence suggests that brain activity at the population level frequently exhibits spatially and temporally structured patterns. The study of these patterns began with detecting plane waves in electrode studies on the cortical surface.[20] Spatiotemporal patterns, including travelling waves,[21][22][23] spiral waves (that rotate around a central point),[24][25][26] source and sink patterns (which expand or contract respectively) and saddle patterns[24][27](which are formed by the interaction of multiple waves) are now acknowledged as common in brain activity at various scales.[28][8] These waves exhibit substantial variation depending on the recording methods and brain regions being studied. Some studies show waves moving cohesively across several cortical areas,[29] while others report waves compressing and reflecting at regional boundaries.[30] The dynamics of these waves are also dependent on recording modality; propagation speed measurements in EEG alpha oscillations have demonstrated speeds on the order of 10 m/s,[31] whereas waves in voltage-sensitive-dye (VSD) imaging of cortical slices demonstrate speeds in the range of 0.03 m/s.[32]

Wave patterns have frequently been observed in studies of sensory responses. Changes in stimuli can initiate activation patterns that emanate from locations within the relevant sensory cortex, subsequently propagating across the cortex and retracting toward a designated sink location. Such source and sink dynamics have been recorded using VSD imaging in the rodent sensorimotor cortex following whisker stimulation and limb stimulation.[27][9] This suggests that these patterns may be primarily governed by the brain's structural connections and, in the absence of explicit stimuli, spontaneous activity might replicate previous sensory patterns to fulfil a cognitive function.

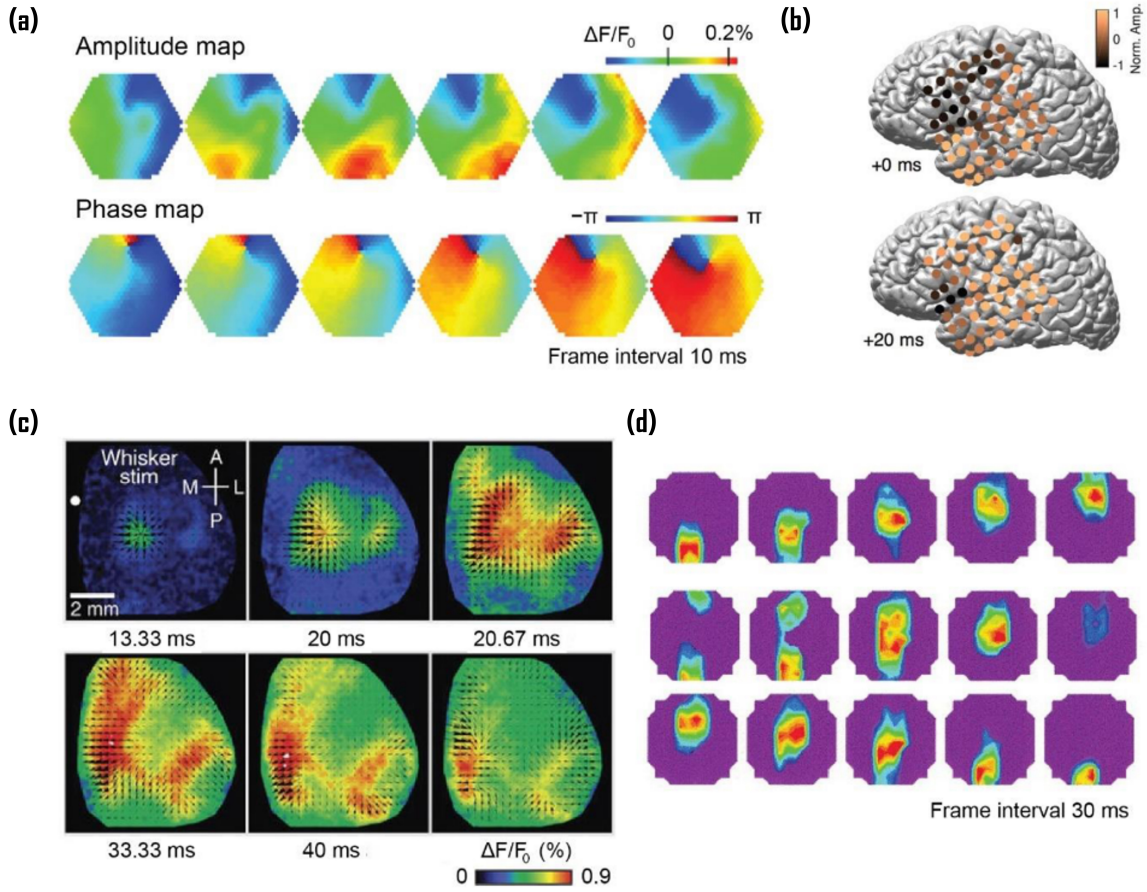

Figure 1.1: Spatiotemporal patterns in neural recordings (a) Spiral wave in voltage-sensitive-dye imaging from [25]. **(b)**Spiral wave in human sleep spindles measured by intracranial ECoG from [26] **(c)** Whisker-stimulus-evoked source and sink pattern in rat VSD imaging from [27] **(d)** Interacting patterns in rat cortical slice from VSD imaging. Rows representing different trials from [33]

The functional significance of these spatiotemporal patterns remains a focal point of ongoing investigation. Research has demonstrated that rather than manifesting randomly, propagating patterns in spontaneous activity adhere to recurrent temporal motifs.[24][27] These patterns are hypothesised to enhance the transfer of information between different regions of the brain and to support distributed dynamical computation.[22][34][35] In the context of sensory cortices, specific stimuli are known to induce consistent propagating patterns,[8][22][29][30][36] with the characteristics of these waves correlating to distinct stimulus attributes. For example, in the motor

and visual cortices, the phase and amplitude of travelling waves are associated with the location of reach targets and the magnitude of rapid eye movement, respectively.[34][37] Additionally, the direction of wave propagation in the visual cortex is influenced by the orientation of visual movement.[38]

These findings underscore the importance of detecting and analysing these patterns to elucidate the principled dynamics of neural population activity and to comprehend the operational mechanisms of neural circuits.[27][36][39]

1.2.2 Temporally correlated functional networks

Another lens for studying spatiotemporal patterns frequently adopted by neuroscientists is the study of functional networks through fMRI. This framework has yielded significant findings and provided profound insights into the brain's integrated and dynamic architecture. By analysing temporal correlations in the BOLD signal, researchers have mapped out large-scale networks crucial for various cognitive functions. This analytical approach, focusing on correlations persisting over several minutes, has identified multiple functional networks, including the somatomotor network, attention network, the default mode network, and more, as shown in figure 1.2. By examining the temporal correlations in the BOLD signal, this anal-

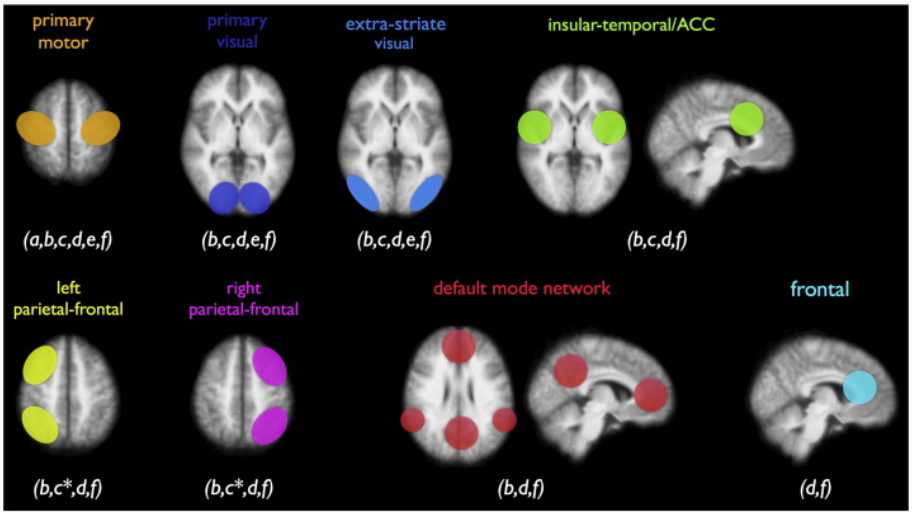

Figure 1.2: Functional networks of the human brain: This figure shows consistently reported resting-state networks across a variety of studies and methods, including the primary sensorimotor network, the primary visual and extra-striate visual network, the left and right parietal-frontal networks and the default mode network. Adapted from [40].

ysis reveals how disparate regions of the brain co-activate during different cognitive states and tasks, providing insights into the brain's integrated operations.[41][42] The interactions between these functional networks play a critical role in cognitive tasks. These interactions are modulated by neural activity, which coordinates across different brain regions to facilitate complex cognitive functions. Understanding these network dynamics is essential for revealing how cognitive processes are supported by the brain's structural and functional architecture.[43][44]

Recent technological advancements have significantly enhanced the detection of

functional networks through fMRI. Improvements in imaging technology, data processing algorithms, and analytical techniques have allowed for more detailed and accurate mappings of these networks, furthering our understanding of the brain's organizational properties at a macroscale level.[14][45] The accuracy of functional interpretations from fMRI data is heavily dependent on the ability to distinguish true neural signals from background noise. The frequency spectrum of fMRI signals has been intensely studied in the literature. It has been established that cardiac and respiratory rates oscillating around 1–2 Hz and 0.3 Hz [46] have a significant effect in aliasing fMRI data, which is typically collected with a Nyquist frequency of 0.25 Hz. There is also an effect known as 'low-frequency drifts' observed in time series fMRI data, which has been attributed to scanner instabilities.[47] With the high and low-frequency components of this signal heavily influenced by the noise, it is standard practice to apply a band-pass filter to the BOLD signal, restricting the frequency range to what is known as the infra-slow range (0.01-0.1Hz) to reduce noise contamination. Nonetheless, noise can still affect the signal within this filtered range. In many studies, spatial filtering is further applied to address the noise concerns in fMRI data analysis. This method encompasses a variety of techniques designed to enhance the spatial resolution of images, thus improving the detection of true neural signals while suppressing noise elements that do not correspond to neuronal activity. Techniques such as Gaussian smoothing, which applies a convolutional kernel to the spatial data, can blur excessive noise while preserving the signal from active brain regions. This smoothing improves the signal-to-noise ratio (SNR) by averaging out voxel intensity values over a spatial extent, thereby reducing the variability attributable to random noise. The effectiveness of these spatial filtering techniques in enhancing SNR has been demonstrated in both task-based and resting-state fMRI studies; Gaussian smoothing with filter widths ranging from 2 to 15 mm has improved the signal-to-noise ratio and facilitated more significant overlap in group-level statistical analyses. [48] This indicates the value of conducting analyses across a spectrum of filter sizes, revealing the significant influence of filter width on functional interpretations. Such methodologies underscore the critical nature of meticulous scale selection in fMRI studies, ensuring the accurate detection and interpretation of brain activities.[48]

1.3 Interdisciplinary relevance of spiral waves

Transitioning from their appearances in brain activity this section serves to spotlight spiral waves and the insights they have provided in complex systems from optics to cardiology, frequently playing a crucial role in the organisation of spatiotemporal dynamics.

1.3.1 Identification of spiral waves

Invariably located at the centres of spiral wave patterns in excitable systems are topological defects.[49] The mathematical description of these singular points of discontinuity relate to the notion of order-parameter fields.[50] A well-studied example of such fields is the phase field $\phi(\vec{x}, t)$ in which spiral wave centres are referred to as phase singularities, where phase values vary continuously from $-\pi$ to π.[49][51] The definition of phase singularities is thus quite naturally given by an index that

counts the winding number of the phase field around the singular point, known as the topological charge. This can be calculated through the spatial gradient of the phase $\nabla\phi(\vec{x},t) = \frac{\partial}{\partial x}\phi + \frac{\partial}{\partial y}\phi$

$$n = \frac{1}{2\pi}\oint_{\Gamma}\nabla\phi(\vec{x},t) \tag{1.1}$$

Phase singularities have increasingly been recognized in biological processes, notably in the concentration patterns of signalling proteins within cell membranes and cardiac tissues.[52][53] Their identification through topological charges has been vital in cardiology to understand cardiac cell synchronization and its implications for ventricular fibrillation—a critical cardiac disturbance characterised by rapid, irregular heart rhythms. These insights are crucial for exploring disruptions in cell synchronization that could precipitate severe cardiac events, such as sudden cardiac death.[53][54]

A recent study has demonstrated the robustness of phase singularity detection in fMRI signals.[55] The existence of these singularities has been unequivocally confirmed via Fourier-based spatiotemporal phase randomisation. This method, while maintaining the original signal's spatial and temporal autocorrelations, modifies phase structures to verify the authenticity of detected spirals, thereby excluding data processing artifacts. The effectiveness of this technique is further bolstered by comparing phase maps of filtered signals against those generated by surrogate data, using a similarity index to confirm the significance of the spirals.[55] Moreover, spiral wave detection is insensitive to the flattening procedures in fMRI data processing. This is evident through consistently detecting spirals across various cortical areas, regardless of mapping techniques such as those involving sulcal depth. This suggests that the structural and functional integrity of spiral waves in the brain remains intact despite potential alterations in cortical surface representations. The centres of spiral waves follow super-diffusive trajectories, indicating their dynamic nature and suggesting a computational role in brain activity. This super-diffusion, characterised by a more extensive and faster movement than typical diffusion processes like Brownian motion, hints at active transport mechanisms in the brain that are critical for rapid neural network reconfiguration in response to cognitive demands.[55]

Exploring these spiral waves in relation to motor tasks could reveal significant insights into how these spirals influence or coordinate large-scale neural dynamics. Such research could expose new dimensions of the neural foundations of movement and motor planning, enriching our understanding of the intricate interplay between cognitive functions and motor processes. This exploration promises to enhance our comprehension of brain dynamics orchestrated across various tasks, providing a detailed perspective on brain function and interaction.

1.3.2 Modelling spiral waves

Phase singularity modelling is pivotal in understanding the self-organizing behaviours observed across diverse disciplines due to its fundamental association with spiral waves, often described as defect-mediated turbulence or spiral defect chaos.[56] This dynamic phenomenon, characterized by the periodic emergence and dissolution of

phase singularities, underscores complex behaviours in various systems, thereby establishing a crucial framework for studying chaotic dynamics. In fields like nonlinear optics and laser physics, phase singularities offer insightful "spiral solutions" to the complex Ginzburg-Landau equation. This pivotal reaction-diffusion equation guides wave propagation in stochastic media.[57][58] These singularities qualitatively influence their immediate surroundings, effectively organizing the wave field through their strategic configurations. Termed a "singular skeleton" in optics, this structure provides a compact yet comprehensive representation of wave dynamics, significantly enhancing the efficiency of optical information encoding and retrieval. [59] [60]

Methodologies for constructing phase maps from singular configurations are thus relevant in many fields. This singularity-based modelling has been explored previously for two-dimensional manifolds, demonstrating the feasibility of these maps only when specific topological conditions of the medium are met. In the literature, numerical strategies for mapping complex geometries—from plain surfaces to intricate geometries such as cardiac tissues—support significant advancements in computational experiments, particularly in cardiac electrophysiology. This methodological progress provides a solid mathematical foundation for further exploring and modelling complex systems where phase singularities are integral.[61][62]

These studies highlight the versatility of phase singularity models in linking theoretical constructs to practical outcomes in various scientific fields. The potential for extending these models into neuroscience, where understanding complex neural dynamics is critical, suggests a promising avenue for future research. This interdisciplinary approach deepens our comprehension of physical phenomena and facilitates innovative solutions to longstanding problems in neuroscience.

Chapter 2

Identifying Spiral Waves in Motor Task fMRI

The observation of spiral waves in filtered fMRI signals represents a bridge for studying the spatiotemporal dynamics of the brain, as one may study other excitable systems such as turbulent flows. The Spiral waves studied here indicate rotational structure within spatiotemporal oscillations in the motor cortex. In section 2.1 the pre-processing required for the analysis of fMRI signals is outlined, followed by the robust quantification of the rotational structures using mathematical definitions. Section 2.2 follows the properties of the identified spiral waves, determining unique qualities of their response to external input in the form of the onset of motor tasks. This section reveals numerical experiments for characterising spiral waves and the dynamic manner in which they emerge, providing insights into their propagation and adaptive capabilities.

2.1 Data pre-processing pipeline

2.1.1 Minimally pre-processed fMRI data

The data examined in this thesis was collected by the Human Connectome Project (HCP) and made available in their 1200 subjects data release[63]. This data collates the MRI scans from 1206 healthy young adult participants. We examined a random sample of 150 of these participants. The HCP is thorough in its scanning protocols and addresses key concerns with fMRI and the data collection process through consistent methodology. Head motion is addressed by an optical motion tracking camera system with the ability to interrupt recording due to rapid head movement or suprathreshold displacement of the head's position. Motor task fMRI data was collected in block format where each block of a movement type lasted 12 seconds, consisting of 10 movements, and was preceded by a visual cue (3 s). Each recording session captures 13 blocks, with two blocks for each movement of the left hand, right hand, left foot, right foot, and tongue. In addition, there are three 15-second fixation blocks per session, and each task is computed against this fixation baseline. The HCP did not provide any in-scanner performance metrics for these tasks, such as reaction time or accuracy, due to the fact that there is no subject choice response for the motor task. Finally, the data undergo structural preprocessing to

align and normalise the structural MR images to standard templates, which assists in accurate brain region identification and quality control measures are rigorously applied, involving both automated and manual reviews to ensure data consistency and reliability

2.1.2 Details of the pre-processing pipeline

From the acquired data, we take the raw signals in 2mm standard CIFTI grayordinate space, composed of 32,000 cortical voxels for each subject and the corresponding brain surface structural files, as input. The left and right hemispheres of the cortex are then extracted separately using the surface structural files. Each voxel comprising the cortex is temporally filtered to the infraslow (0.01-0.1 Hz) frequency range, and the resulting image of the cortex is then filtered by a difference of Gaussian band-pass filter to a spatial scale presented in previous studies as capturing spiral wave structures with the maximal functional correlates.[55] Finally, each hemisphere of the cortex is projected onto its corresponding flattened cortical maps to simplify subsequent analysis, the resulting filtered signal and flattened maps are shown in figure 2.1. To capture the dynamics of the fMRI signal, one can examine the analytical phase of the narrow-band time series data from each recorded cortex voxel utilising the Hilbert transform (via MATLAB's 'Hilbert' function). This transform is suited explicitly for narrow-band signals because it relies on the assumption that the signal's amplitude and phase vary slowly compared to the carrier frequency, which is the primary characteristic of narrow-band signals. This detailed phase information is vital for understanding how different brain regions interact over time, offering insights into the neural basis of cognitive functions and potential disruptions in pathological states. The analytical phase values were then displayed on the 2D-flattened representation of the cortical area, creating a phase map $\psi(x,t)$. Based on this phase map we may calculate the spatial derivative through a central finite difference approximation, discard derivatives at the boundary of the cortex, and calculate angle differences using circular subtraction. We then analyse the chaotic evolution of the phase map by considering the role of wave propagation in this turbulent system. This wave propagation can be understood through the definition of the flattened cortical representation's spatial phase gradient as the wave vector field.

$$k_\phi(x,t) = \nabla\psi(x,t)$$

Focused particularly on spiral waves, we identified phase singularities, which are invariably located at the centres of spiral wave patterns. As described in section 1.1, phase singularities are defined by the winding number of a closed contour surrounding the singularity's location, also known as their topological charge. Hence, by considering equation 1.1, one may evaluate the topological charge n for a small circular path Γ of infinitesimal radius a using the curl of the wave vector field. [53]

$$(\nabla \times k_\psi) \cdot \hat{z} \equiv \lim_{a\to 0} \frac{1}{\pi a^2} \oint_\Gamma k_\psi \cdot d\mathbf{l} \tag{2.1}$$

The curl of the wave vector field may be measured using the curl function in MATLAB. Thus, for each point, we evaluated the topological charge simultaneously, dramatically improving the efficiency of this detection method. A vector field's

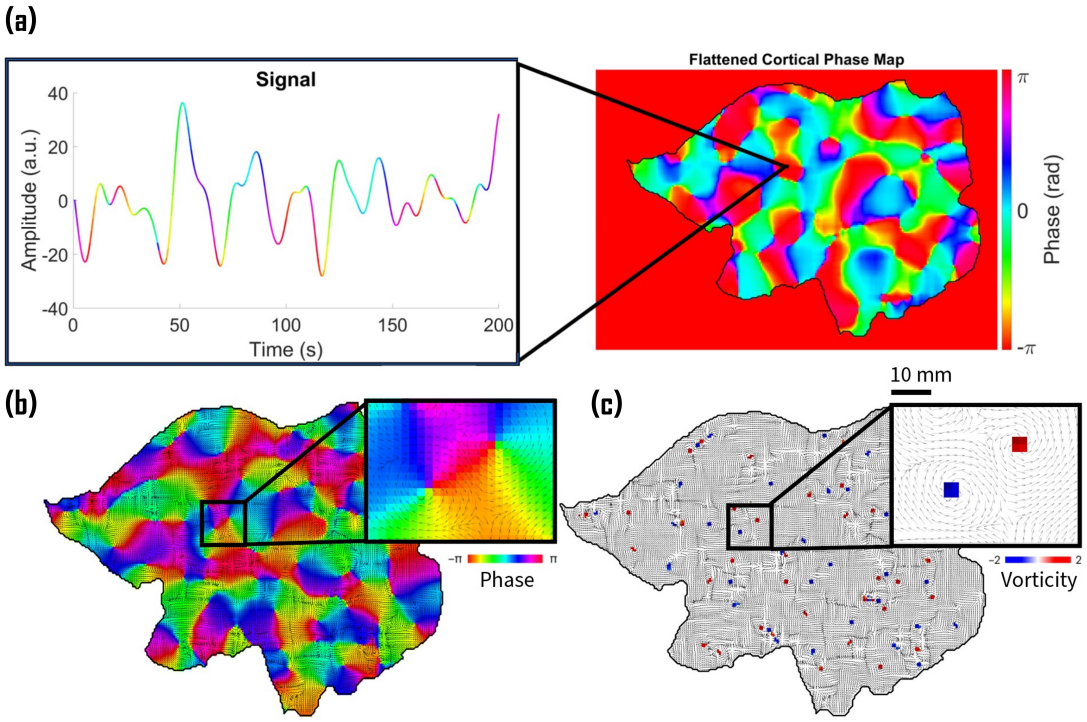

Figure 2.1: Detection of spiral waves in filtered fMRI signals **(a)** The spatially and temporally filtered fMRI signal from a single voxel within the flattened left hemisphere demonstrating the relationship between the analytical phase as computed by the Hilbert transform and the amplitude of the fMRI signal. **(b)** Two Phase singularities are shown amongst the overall dynamics of the phase field **(c)** Vorticity thresholded phase singularities within the wave vector field computed from the spatial gradient of the phase map.

curl is also called the field's vorticity. By thresholding the vorticity above 1 or below –1, one directly captures whether the spiral wave is rotating anti-clockwise or clockwise, respectively. This threshold ensures that identified points are surrounded by all phase values, satisfying the definition of a phase singularity outlined in the literature.

2.2 Spiral Wave Properties

Just like the wave patterns present in the equilibrium and non-equilibrium systems discussed in the literature. Individual spiral waves evolve and interact, shaping the ongoing cortical dynamics as they form and dissipate across the infra-slow temporal scale. Directly retrieving properties of this chaotic evolution was non-trivial but critical for understanding the mechanisms underlying wave propagation across the cortex.

2.2.1 Spiral wave response variability

To explore the response of spiral waves to the onset of motor movement in task conditions, one can distinguish between spiral waves generated before the task onset and spiral waves generated during the task. Previous studies [55] have demonstrated

that spiral wave location and rotation encode cognitive task information; hence, to describe the response of spiral waves for each subject, one focuses on these two aspects. Based on the potential for spiral waves and their centres to traverse the cortex, it was hypothesised that the brain would use previously existing structures of brain activity in an energy-efficient manner to coordinate motor movement. This hypothesis suggests that one should see the majority of spiral waves generated before the task onset and respond to the task by being modulated into a location that would be similar across different subjects. One may implement a density-based clustering algorithm for spiral centre locations to investigate this hypothesis. This algorithm allowed for the overlay of responses for all subjects, identifying particularly dense or frequent regions of spiral wave activity in response to the task onset. As each subject completed a particular movement for two trials sequentially, one should consider the spiral wave response to each trial independently for all subjects.

The DBscan function in Matlab requires a few inputs: Z, ϵ and MinPts. Z is a matrix where each row represents an individual data point, and each column represents a data feature. For our case, Z contained the pre-computed distances between each identified spiral centre location at a particular time frame. Epsilon, the second input, is a scalar that specifies the maximum distance between two points to be considered part of the same cluster. Using the knowledge that spiral cores are approximately 4mm, one can set epsilon to be 10mm, allowing for diversity in the size of the spiral wave one includes while also ensuring that clusters are highly dense. MinPts is the minimum number of points required to form a dense region; this defines how many neighbouring points are needed to start and expand a cluster. This parameter must be optimised to ensure that only the densest clusters at our selected ϵ are examined. Iterating over values of MinPts starting with the densest clusters, DBSCAN is applied, and the number of clusters is identified. This process continues until a minimum of two clusters are identified, ensuring that only the densest clusters are highlighted. The systematic reduction and conditional checking make fine-tuning the clustering process and capturing the most significant dense regions easier. For consistency and to allow time for any delay from fMRI signals, one considered a snapshot of spiral locations from the middle of the task period.

Figure 2.2 demonstrates the spiral wave response in the brain to right-hand movements with some puzzling inconsistencies between the two movement trials. Firstly, it appears that during the first movement trial, an equal proportion of spiral waves with the same rotation gather in frequent locations regardless of whether they were generated before or during the task. This observation seems to counter our hypothesis, suggesting that spiral waves in some subjects are generated at a specific location when needed. However, in other subjects, existing spiral waves may be repurposed and follow the super-diffusive trajectories discussed in the literature to a task-specific location. In the second trial of movements, our hypothesis appears more plausible. The dense regions identified in trial 1 are now more frequently populated by spirals generated before the trial onset, and spiral waves generated during the task, although still present at those locations for some subjects for many, are no longer apparent. The findings here suggest that spiral waves in the brain play a complex and adaptive role in coordinating motor movement. This adaptation could reflect a learning or optimization process in the brain, utilizing established neural pathways more efficiently in subsequent trials. The inconsistency between trials

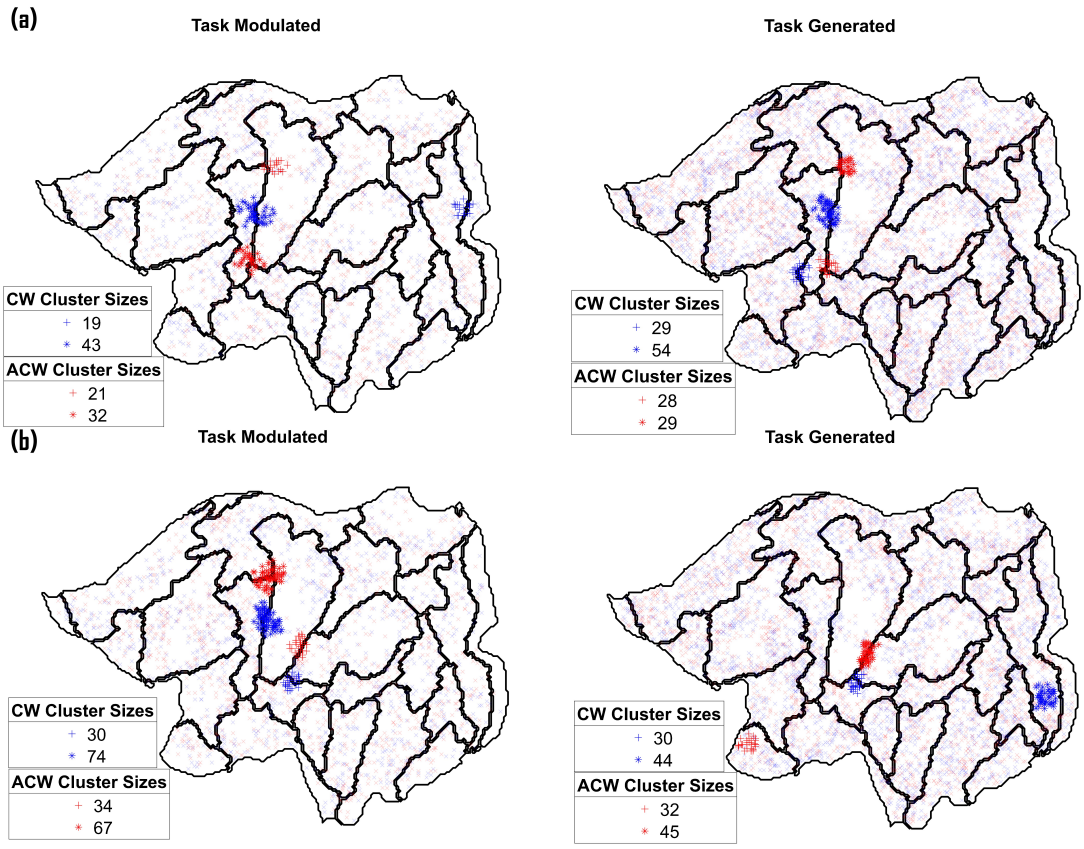

Figure 2.2: Density based clustering: spiral centres at the middle of right hand movement tasks, **(a)** Demonstrates the clustering during the first movement trial with spiral centres originating prior to task onset and spiral centres generated during the task period, **(b)** Demonstrates the clustering during the second movement trial. Transparent crosses capture spiral centres not belonging to any cluster

highlights the brain's dynamic ability to modify its activity patterns, potentially optimizing energy use and response strategies over repeated tasks. While the brain might leverage pre-existing spiral structures, it also retains the flexibility to generate new patterns in response to changing demands, illustrating a balance between energy efficiency and adaptive flexibility. This reorganization could underpin the brain's adaptive responses to varying cognitive demands, reflecting a fundamental aspect of neural processing that contributes to the efficiency of brain functions.

2.2.2 Tracking characteristics of spiral waves

Here we outline a few techniques for evaluating the characteristics of spiral waves such as their formation, trajectory, and stability. For each identified phase singularity, we implemented a k-best global nearest neighbor assignment based on the Jonker–Volgenant algorithm [64] from the Sensor Fusion and Tracking Toolbox in MATLAB to accurately track phase singularities from the frame they were generated to the frame at which they disappear.

Although identifying spiral waves for a single frame is robust, keeping track of the sporadic and non-linear motion of the spiral centres requires careful tuning. Initially,

we complete the phase singularity identification process for two successive frames. We construct a cost matrix based on the distance from each phase singularity in the first frame to all phase singularities in the second frame, taking into account the topological charge of singularities in each frame. The cost matrix contains the cost of each possible assignment, hence the smaller the displacement of a phase singularity from one frame to the next, the more likely it is to be tracked. Our algorithm sets the maximum cost for assignment, which means the maximum distance a phase singularity can move in one time step to be correctly tracked at 20 mm. This parameter ensures that tracked spiral waves are accurately detected while enabling the algorithm to identify the rapid movement of these patterns across the cortex. We use a tracking process to follow spiral waves in all frames of the recording period for each subject. This process returns tracked spirals along with their unique ID. As new spirals are formed, they are saved with the frame in which they were generated. When a spiral is no longer detected after being tracked, we record the number of frames since it was generated as its lifetime. In figure 2.3, we display the distribution of when spiral waves were generated over the entire recording period, indicating the task in which they were generated. We also plot the distribution of the measured spiral wave properties, including the instantaneous propagation speed. We calculate this speed by considering the magnitude of displacement of a spiral wave's centre divided by the temporal resolution of the recording. We express this as $\tilde{v} = |\frac{\Delta v}{\Delta t}|$.

Figure 2.3**(b)** illustrates that most identified spiral waves exhibit short lifetimes with a peak in distribution at less than 3 seconds. However, the average lifespan of spiral waves is 5.4 ± 0.6 s, indicating that there is still a significant portion of spiral waves that persist throughout the recording period. This finding is further backed up by figure 2.3**(a)** showing spiral waves persist beyond the specific tasks that initiated them, indicating their potential role in maintaining cortical readiness and influencing subsequent neural processing. Figure 2.3**(c)** further demonstrates that these patterns are not stationary, with an average propagation speed of 3.1 ± 0.3 mm/s. This quantitative finding suggests a rapid, dynamic rearrangement of cortical activity patterns and, when viewed from the perspective of spiral wave response variability in figure 2.2, illustrates spiral waves as playing an active role in quick transitions between brain states to meet the demands of motor tasks.

In summary, the unified methodology and observations of this chapter demonstrate the transient yet influential nature of these waves in fMRI signals. The insights provided here serve as a foundation for inquiry into the precise functional relevance of these spiral waves.

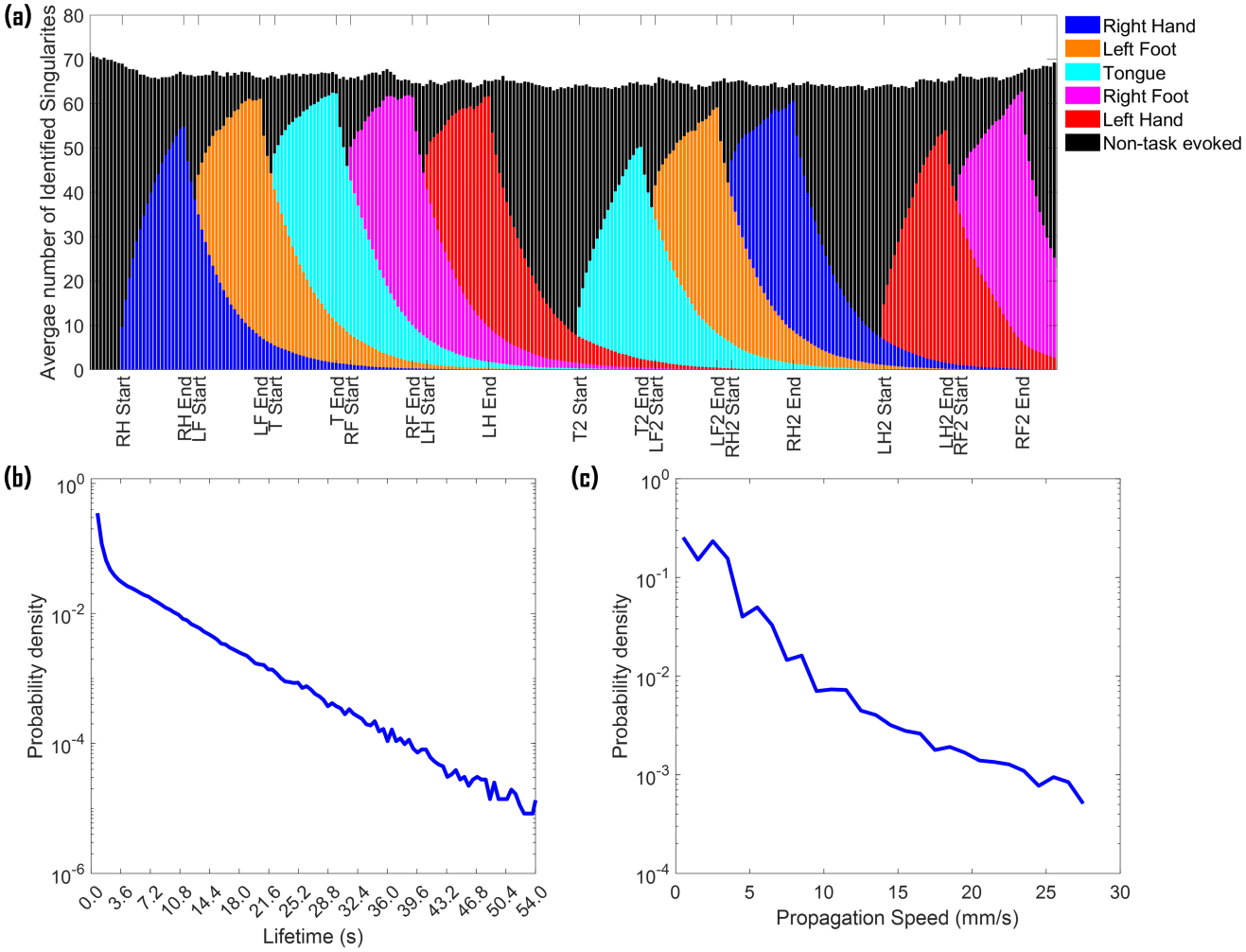

Figure 2.3: Characteristics of spiral waves (a) The total number of identified spiral waves for each frame of the recording, the portion of spiral waves generated within a particular movement task is indicated by the colour scheme. **(b)** The probability density function of spiral wave lifetime for all identified spirals. **(c)**The probability density function of spiral wave propagation speed for all identified spirals.

Chapter 3

Functional Role of Spiral waves

Following the investigation and definitions of spiral waves characteristic of the previous chapter, this chapter delves into the practical implications of spiral wave dynamics within the human brain during motor tasks. By studying the functional roles of these waves, we can gain insights into their universal properties and contributions to system stability and functionality. This comparative perspective not only underscores the significance of spiral dynamics in complex biological systems but also enhances our understanding of how such patterns can be leveraged to interpret cognitive and neural dynamics more effectively. Section 3.1 systematically analyses the task-specific information encoded by the spatial and rotational characteristics of spiral waves, demonstrating the group-level significance of these properties across multiple tasks and then quantifying this significance through classification. Section 3.2 follows this by illustrating the underlying mechanisms that these spiral waves represent.

3.1 Spiral wave task information decoding

3.1.1 Task-specific trial averaged spiral density map

To investigate the amount of task-related information captured by locations and rotations of spiral waves in an individual's neural recording, we computed the task-specific, moment-by-moment spiral distribution maps for each trial - employing a circular boundary with a radius of 10mm around spiral centres as identified in section 2.1. A frame at the mid-point of the task duration was selected for all motor tasks to accommodate the latency associated with fMRI signal responses. Within each task condition, we allocate the spiral wave topological charge (either -1 or 1) to each voxel encompassed by the boundary of a spiral wave. This methodology thus resolves potential ambiguities in spiral distribution by eliminating overlapping regions between spirals with opposing rotational directions. Next, one can compute the average spiral density maps for the combined population by aggregating the instantaneous spiral distributions across all trials and subjects for each voxel. This sum is then divided by the total number of trials associated with a specific task:

$$D_j = \sum_{i=1}^{N} \frac{d_j^i}{N} \tag{3.1}$$

In equation 3.1 D_j represents the population-combined trial-averaged spiral distribution of voxel j, d_J^i represents the moment-by-moment spiral distribution of voxel j from trial i, and N represents the total number of trials across subjects. Considering that clockwise and anticlockwise spirals neutralize each other during the averaging process, the value of a voxel in the averaged spiral density map reflects the additional likelihood of encountering an anticlockwise (or clockwise) spiral over a clockwise (or anticlockwise) spiral, expressed as a percentage of all observations at that voxel. For instance, if voxel j shows a positive value of $D_j = 0.30$, this indicates an additional 30% chance of seeing an anticlockwise spiral at that voxel across all trials and subjects. On the other hand, a negative value of -0.30 signifies an extra 30% chance of seeing a clockwise spiral wave at that voxel.

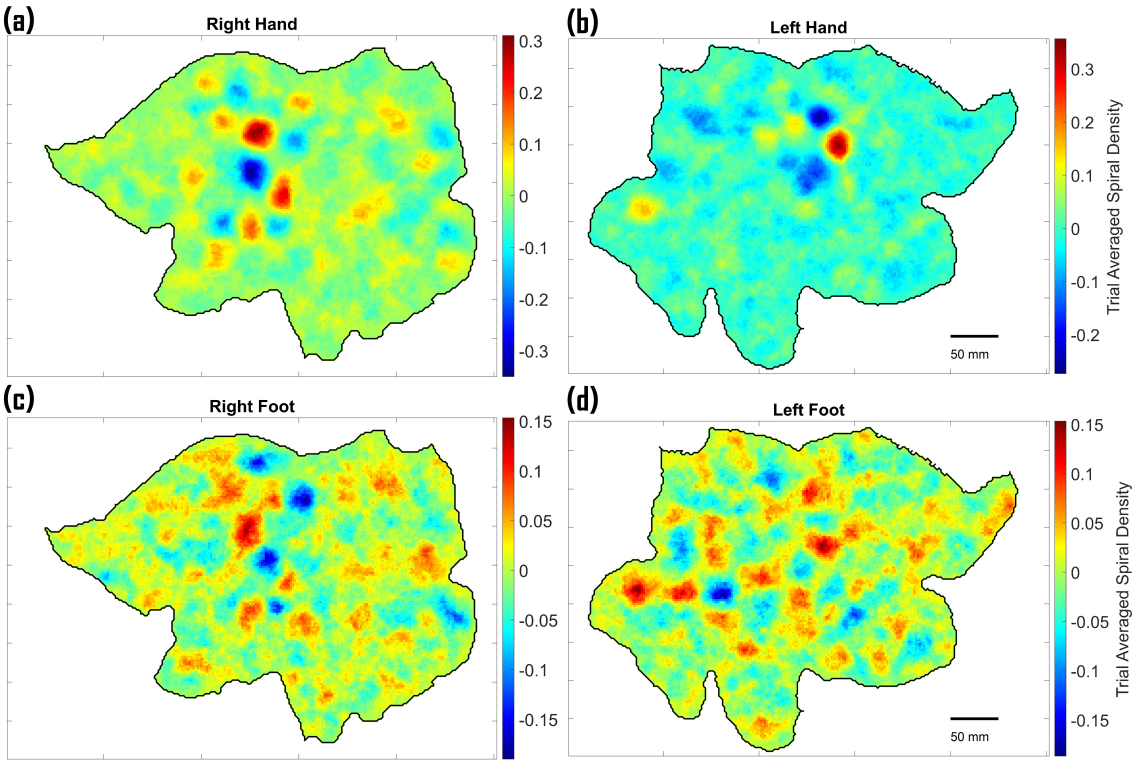

Figure 3.1: Trial-averaged spiral density map during motor tasks cortical regions that are dominated by clockwise or anticlockwise spiral clusters are marked in blue or red colours, respectively. **(a)** Left hemisphere view for right hand movement task. **(b)** Right Hemisphere view for left hand movement task. **(c)** Left hemisphere view for right foot movement task. **(d)** Right hemisphere view for left foot movement task.

The spiral wave densities, as depicted in figure 3.1, appear to be task-specific based on the different distributions observed. The dominance of the spiral wave densities in hand movement tasks may reflect the higher functional resolution essential to the finer motor skills of the hands compared to the feet and the smaller size of the anatomical region related to foot motor function. The occurrence of oppositely rotating spiral waves and their proximity suggest intricate interactions closely linked to each motor function. This arrangement is notably symmetrical across hemi-

spheres for hand movements, which indicates a robust mechanism for maintaining balance and symmetry in motor functions, which is critical for coordinated bilateral movements.

Based on the density maps observed in figure 3.1, spiral waves with specific rotations and locations are consistent across subjects. Moving forward, we analyse the functional significance of these spiral waves by predicting task condition based on the presence of spiral waves in these task-specific locations.

3.1.2 Global spiral-centre-based linear classifier

We need a more robust quantification than simple density maps to determine whether spiral wave location and rotational properties encode the neural processes related to motor tasks. However, from the observations of figure 3.1, one can assume that the spiral centre distributions for the same task condition at any given moment should be more similar to each other than those from different task conditions. This similarity suggests the possibility of comparing an individual's instantaneous spiral centre distribution to the average spiral centre density map for a specific task condition to determine the task condition for that individual. First the instantaneous spiral centre distributions for each subject are identified using the methods outlined in section 2. To distinguish true spirals from false positives, one measures the similarity between their rotational phase structures in filtered and unfiltered signals. A true spiral should have similar rotational phase structures in the spatially filtered signals as those found in the unfiltered signals. This results in a significantly higher similarity measurement than chance, which we determine using a surrogate data null model generated using the same methods as in a previous study. This step was important as it aimed to remove artefactual spiral waves that may have resulted from the filtering processes, ensuring that only true spiral waves are used for the classification.

To construct the classifier, a cross-validation approach was implemented. This involved selecting 80% of the 300 trials completed for each movement task as the training set. The remaining 20% of trials of each movement task were used to test the accuracy of the classifier. The spiral distribution for each trial was obtained by averaging the spiral locations over the middle frames of the trial. The resulting distribution for each trial was then summed and divided by the size of the training set to attain a population-level representation of spiral wave activity during a specific movement. This averaging process resulted in 6 templates of spiral activity during each motor task, with movements of the right limbs being observed through the left hemisphere of the brain and movements of the left limbs being observed through the right hemisphere of the brain. The trials of tongue movements were also included, as tongue movements may originate from both hemispheres of the brain. In the human brain, each hemisphere primarily controls the movements of the opposite side of the body. This is due to the organisation of the brain's neural pathways, where the motor control nerves extend from one hemisphere of the brain and predominantly cross to the opposite side of the body. Thus, movements of the right hand are primarily controlled by the left hemisphere and movements of the left hand by the right hemisphere. This crossing of the pathways ensures that each hemisphere specialises in processing motor functions for the opposite side. Taking

this into consideration, we considered the classifications of different tasks bilaterally.

Considering observations through the different hemispheres separately, the testing data from each movement task were compared against each spiral activity template using Pearson's 2D correlation coefficient.

$$r = \frac{\sum_m \sum_n (I_{mn} - \bar{I})((T_{mn} - \bar{T}))}{\sqrt{(\sum_m \sum_n (I_{mn} - \bar{I})^2)(\sum_m \sum_n (T_{mn} - \bar{T})^2))}} \tag{3.2}$$

The task condition corresponding to the template that shows the greatest similarity to the test spiral centre distribution is identified as the task condition.

The accuracy of our prediction is verified by comparing the predicted task conditions against actual outcomes, classified as either hit or miss. The prediction accuracy is the ratio of hits to total predictions, compared against a chance level classification of 33%. For instance, an accuracy of 45% from 100 predictions indicates 45 hits. This process is iteratively repeated 100 times with shuffled training and testing trials to enhance prediction reliability, with the overall prediction accuracy calculated as the mean of accuracies from all iterations. This thorough validation process ensures the accuracy and reliability of our predictions.

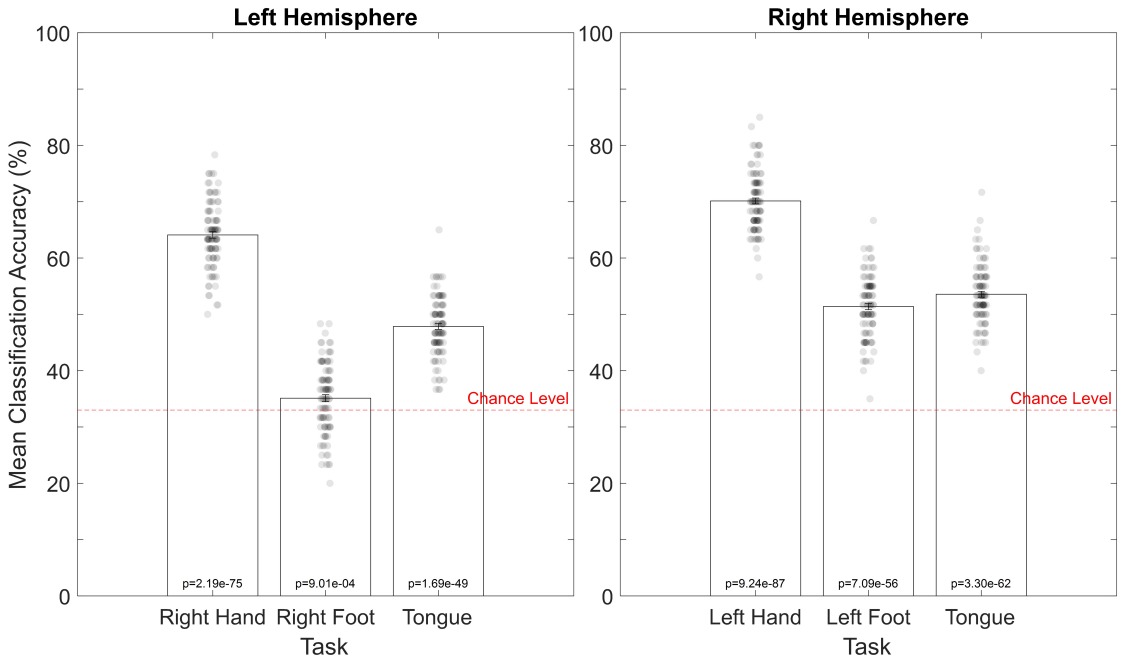

Figure 3.2: Iteration-averaged classification accuracy Results for six motor task conditions, calculated for 150 subjects. The error bars (partially obscured) denote mean±s.e.m., n=100 iterations. The red dotted lines show the chance level (33%). One-sampled t-test shows all classification accuracies are significantly higher than chance (P<0.001, two-tailed). The black dots mark the classification accuracy of each iteration.

In the left hemisphere, the average decoding accuracy for the right-hand task was approximately 64.1%, which was the highest among the tasks analysed in this hemisphere. This suggests a consistent neural representation of right-hand movements

by spiral waves at the dense locations highlighted in figure 3.1. Conversely, the right foot task exhibited a lower decoding accuracy of 35.1%, indicating a less distinct neural pattern, possibly due to the lower motor or sensory specificity of the dense regions in figure 3.1 for foot movements. The tongue task showed a moderate decoding accuracy of 47.8%, reflecting a specialised but less distinct pattern compared to hand movements.

Comparatively, the right hemisphere demonstrated generally higher decoding accuracies. The left-hand task presented the highest accuracy at 70.1%, suggesting a strong lateralisation effect, where the right hemisphere exhibits more pronounced neural activity. The left foot and tongue tasks in the right hemisphere also showed higher accuracies (51.4% and 53.6%, respectively) than their counterparts in the left hemisphere, aligning with the notion of hemispheric specialisation for motor control.

3.2 Wave propagation in the motor cortex

3.2.1 Subject averaged wave vector fields

Studying wave vector fields across many subjects performing the same motor tasks can provide a robust framework for understanding commonalities and variances in motor function. For this purpose, one considered averages of the wave vector field in its complex representation $k_{mn}(x,t) = u_{mn} + i \cdot v_{mn}$, across all trials for right and left limb movements. The magnitudes of averaged vectors at each position were then z-scored according to standard deviations above the average magnitude of all vectors. The average direction was also calculated by considering the angle of the complex representation of averaged vectors. These mean fields are demonstrated in figure 3.3.

Analysing the mean wave vector fields shown in figure 3.3 provides a dynamic view of how brain regions communicate during motor tasks. The direction of the vectors within these fields, indicated by streamlines in the figure, is particularly relevant in understanding how motor commands are initiated and processed. For instance, during a motor task, activation appears to begin in the primary motor cortex and propagate to other regions involved in motor control, such as the pre-motor cortex and supplementary motor area. Specifically, the direction of these vectors may indicate sequential activation, showing the functional neural pathways utilised during the task execution. Figure 3.3 shows that the direction of the wave propagation through these regions is highly consistent, as demonstrated by the polar distributions. Specifically, this consistency is apparent at the locations of high vector magnitude, as indicated by the z-scores. The magnitude of these vectors indicates the wave propagation speed and may also have significant neurophysiological implications. Something particularly interesting about the regions of high vector magnitude in figure 3.3 is the apparent similarity in location to the dense regions of oppositely charged spiral wave centres identified in figure 3.1.

Previous studies on these spiral waves have portrayed this task-specific activity flow on a larger scale by focusing on arrays of many spiral waves.[55] Here, we demonstrate a novel methodology to identify the so-called “gates” formed by pairs of interacting spirals described in this flexibly coordinated activity flow. This method-

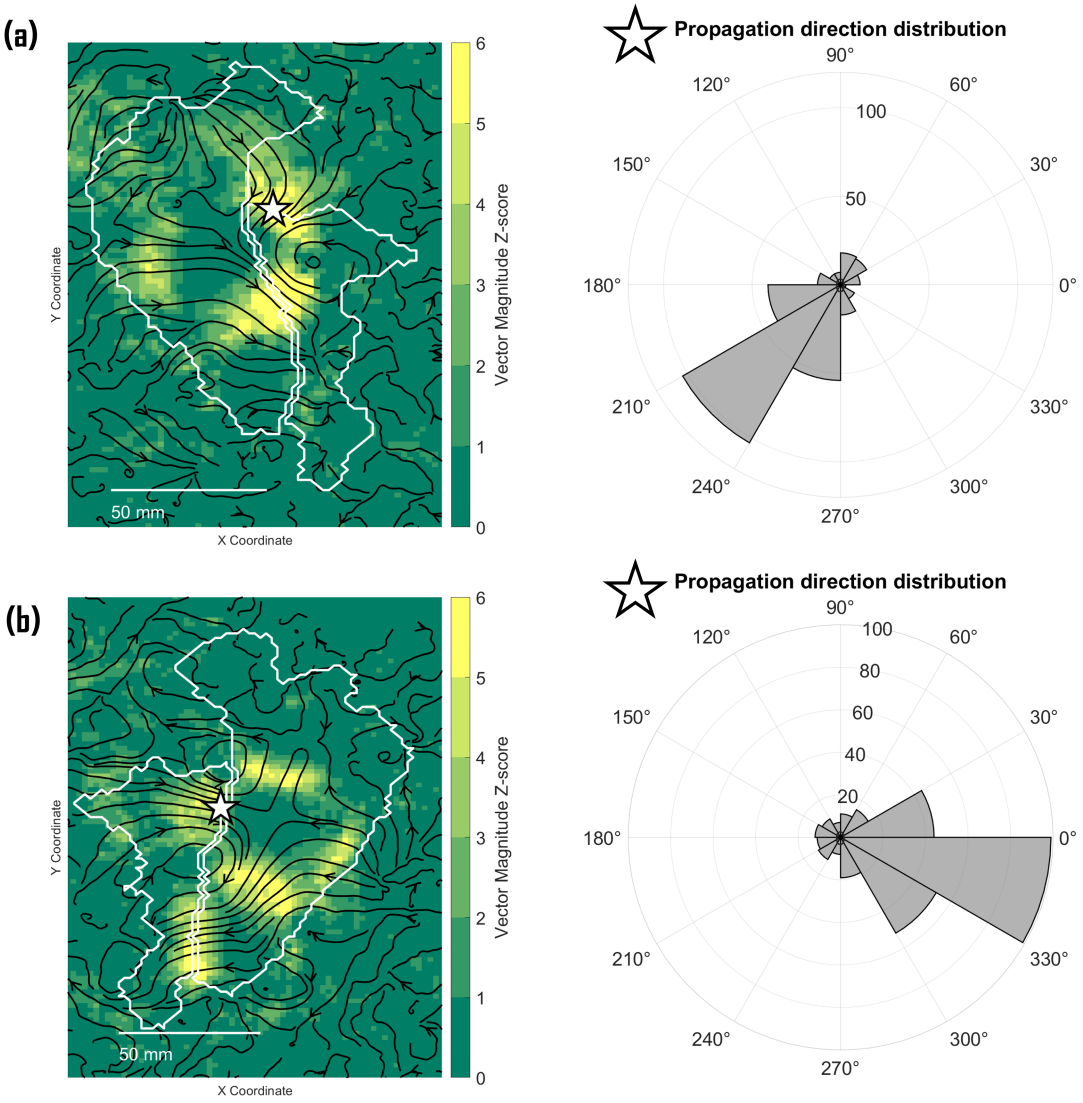

Figure 3.3: Subject averaged wave vector field averaged streamlines overlaid on the vector magnitude z-scores in the motor and pre-motor cortex outlined in white, a) displays results from the right hemisphere of the brain recorded at the end of the left hand trials. b) displays results from the left hemisphere of the brain recorded at the end of the right hand trials. Polar Histograms display the distribution of propagation directions at the starred location for every trial.

ology is further verified by the apparent hubs of activity surrounded by these high-magnitude flows, with these "gates" coordinating both an in-take and out-take of neural activity. This phenomenon points to an underlying neural mechanism potentially related to the functional networks previously studied in fMRI data. Spiral wave interactions may serve as the dynamic process that makes communication between these functional networks possible. It may only be possible to recognise this using this interpretation of moment-by-moment spatiotemporal flow that is not typically captured by the correlation-based analysis frequently used for understanding this data.

This comprehensive analysis of the functional significance of spiral waves in motor tasks has demonstrated how these intricate wave patterns encode task-specific in-

formation crucial for understanding brain motor control and cognitive processing. The detailed exploration of spiral wave densities and the application of classification techniques have not only highlighted their consistent presence across subjects but have also emphasised their predictive power in decoding motor tasks. These findings underscore the critical role that spiral dynamics play in neural coordination and information processing, akin to their functions in other complex biological systems such as the heart. As we move forward, it becomes increasingly clear that an advanced modelling approach is necessary to exploit the potential of spiral waves fully. Similar to their function in other complex systems, such models would allow us to simulate and manipulate the complex spiral wave interactions under various conditions, providing deeper insights into their mechanisms and implications for brain functionality. By integrating this chapter's empirical data and analytical insights with dynamic models that can capture the complex interactions of spiral waves, we can open new avenues for predicting and potentially modulating brain activity.

Chapter 4

Mathematical Modelling of Spiral Wave Interactions

Results thus far have established spiral waves in fMRI phase signals as unique, dynamic features of brain activity, maintaining functional significance through the activity they coordinate. This chapter, through constructing and refining a model based on the configuration of phase singularities, aims to bridge the modelling approaches of complex physical systems with practical neuroscience applications. The development of these models enhances our comprehension of the mechanical roles these spiral waves play in the coordination of neural functions. Section 4.1 follows the construction of a solution through advanced computational methods. Section 4.2 illustrates the validity of the approach through various measures of similarity to actual cortical data.

4.1 Reconstruction of 2D phase maps from the configuration of phase singularities

If one considers a phase map to be determined by the additive effects of multiple phase singularities and their associated spiral waves, we can construct a model that performs this operation, taking the location and topological charge of phase singularities as input. By defining a grid of X coordinates and Y coordinates in MATLAB, it is possible to define a phase singularity at a point (x_0, y_0) using two more arrays that capture the displacement from the point along each axis $X_{rel} = X - x_0$ $Y_{rel} = Y - y_0$ and taking the inverse tan of these displacement vectors using the atan2 function in MATLAB. This operation produces a continuous angular field with a discontinuity at the vortex centre, which is the signature of a phase singularity, as outlined in Chapter 1. Hence, let ϕ represent the angular field computed. Following this operation, ϕ only depends on the location of the phase singularity. The rotation of this angular field may be reversed by simply considering its negative; thus, for one phase singularity with topological charge m, the complete structure of the surrounding phase map is captured by $m\phi$. However, this will only initialize the spiral wave associated with this phase singularity at a rotational phase of 0 degrees with respect to the x-axis. As this is not physical, one must take further inputs to the model. Specifically, one must define a rotational phase of the spiral wave associated

with each phase singularity. Now one has for a single phase singularity at location (x_0, y_0) with topological charge m and spiral wave rotational phase ω an array that describes the structure of the surrounding phase map given by:

$$\hat{\phi} \equiv \text{mod}(m(\phi(x_0, y_0) - \omega), 2\pi) - \pi$$

where the modulo operation and π subtraction are done such that $\hat{\phi} \in [-\pi, \pi]$.

4.1.1 Measuring spiral wave rotational phase

Measuring the rotational phase of spiral waves is not trivial. We must first identify the spiral wavefront related to an identified phase singularity. This identification can be achieved by considering the phase map in a small radius surrounding a phase singularity. Using Sobel convolutional kernels, one can perform a convolution on the phase map and detect discontinuous jumps in the phase from $-\pi$ to π. The kernels used for this work are given by equation 4.1

$$k_x = \begin{bmatrix} -0.5 & 0 & 0.5 \\ -1 & 0 & 1 \\ -0.5 & 0 & 0.5 \end{bmatrix} k_y = \begin{bmatrix} 0.5 & 1 & 0.5 \\ 0 & 0 & 0 \\ -0.5 & -1 & -0.5 \end{bmatrix} \tag{4.1}$$

Thus, using the conv2 MATLAB function to perform a 2D convolution on the whole phase map, we identify the phase transitions by thresholding the magnitude of the convolutions by 2π, defining all wavefronts within the original phase map. Focusing on the spiral waves, we then use the small radii around phase singularities to capture the points defining a wavefront for the spiral wave. From this set of points, we must then determine the angle of rotation for the spiral wave. By establishing polar coordinates in the regime of the small radii surrounding the spiral wave, with the origin set at the phase singularity, one can define the angular position of each point that makes up the wavefront. However, some confounding factors arise from this method. Often, phase singularities are spatially separated by a distance less than the small radii we must use to outline the spiral wave, meaning a set of points may be influenced by a wavefront belonging to a different spiral wave. Unfortunately, there are no automatic detection methods for this edge case, so one must carefully use summary statistics to define the rotation angle for each spiral wave.

For this reason, we define the spiral wave rotation by taking the median angular position of each point within the small radius surrounding a phase singularity. The presence of outliers has a lesser impact on this statistic as compared to the mean, so effects from other wavefronts are mitigated. Figure 4.1(b) demonstrates the results from this procedure, with the small radii around phase singularities and lines within those radii demonstrating the angle of the spiral wavefront.

4.1.2 Gradient-based optimization of radial influence

Following the model's construction, note that each array's additive effects ($\hat{\phi}$) has an unconstrained influence on the entirety of the modelled phase map. This leads to unwanted distortion from spiral waves in what should be completely separate cognitive domains. Such interference markedly diminishes the model's precision in capturing the true phase dynamics. This observation underscores the necessity for

introducing a parameter to quantify the radial influence exerted by each phase singularity. Unlike the spiral wave rotational angle, this parameter is not readily available from the actual data. So, one must consider an approximate envelope for this radial influence; for this purpose, we define a Gaussian function that constrains the influence of the array and its additive effects on the modelled phase map described by equation 4.2.

$$\begin{aligned} X_{mn} &= \sum_i e^{-\frac{r_i^2}{2\sigma_i^2}} \cos(\hat{\phi}_{mn}(x_i, y_i, \omega_i)) \\ Y_{mn} &= \sum_i e^{-\frac{r_i^2}{2\sigma_i^2}} \sin(\hat{\phi}_{mn}(x_i, y_i, \omega_i)) \end{aligned} \tag{4.2}$$

Hence, the final model for the phase map of the cortex is given by equation 4.3.

$$\psi_{mn} = \arg(X_{mn} + iY_{mn}) \tag{4.3}$$

In equation 4.3 $\hat{\phi}_{mn}(x_0, y_0, \omega_i)$ is the constructed phase singularity array for a spiral wave at point (x_i, y_i) with rotational phase ω_i. The Gaussian function computes a 2D envelope using the array r_i containing the distance from each point in the array to the phase singularity and σ_i the Gaussian envelope width. This width represents the FWHM of the phase structure surrounding a singularity and thus effectively determines the radial influence of the phase singularity.

While the proposed method allows for control over the parameter σ and the limitation of the additive influence of each array on the overall modelled phase map, the challenge lies in the absence of a readily available method to determine this parameter from the data. This, being the only undefined parameter, presents a complex optimization problem. To address this, we suggest treating this as an optimization problem over all the σ_i. A practical approach involves employing a gradient-descent algorithm aimed at maximizing the 2D correlation between the modeled phase map and the empirical phase map. We initiate the process by setting a uniform radial influence of 10 for all σ values. Subsequently, in each iteration, we reduce the radial influence by 1 unit for each σ and compute the change in the 2D correlation using equation 3.2. Using this change, we approximate the derivative of the cost function with respect to a particular sigma using Euler's method. Based on these gradients, we update each sigma with an appropriate learning rate parameter and repeat the procedure for 200 iterations. The convergence of the cost function over these 200 iterations is demonstrated in figure 4.1.

4.2 Model Validation and Limitations

Once the parameters have been determined through measurement and optimisation, the model demonstrates significant precision in recreating brain states during all motor tasks. In figure 4.2, we selected a frame 3 seconds before the end of each task and produced the model following the procedures outlined above. In all tasks, the model was optimised to a 2D correlation coefficient exceeding 0.71, meaning that the model excels in capturing more than half the variance of the actual phase information from the data. Figure 4.2 also demonstrates the absolute error between

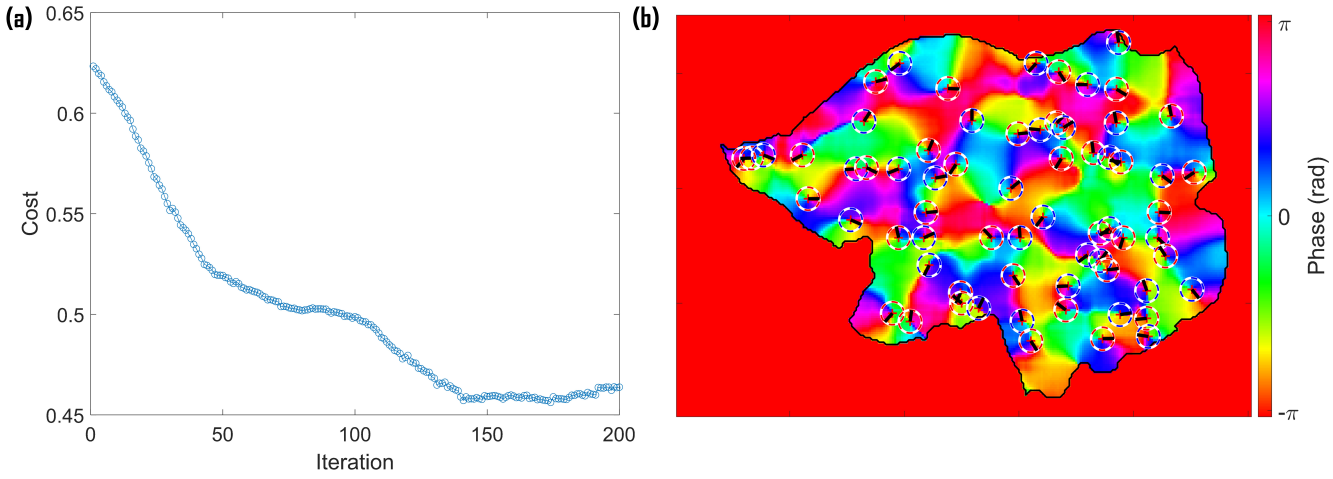

Figure 4.1: Parameter optimization and measurement visualisation (a) Illustrates the convergence of the cost function over 200 iterations of the gradient descent algorithm used to optimize the Gaussian envelope width parameters (σ_i). The cost decreases significantly, indicating an improvement in the fit of the model to the empirical data. **(b)** Indicates the measurement of spiral wave rotational angle using circles surrounding phase singularities, circle colour indicates clockwise (blue) and anticlockwise (red) spiral waves, lines indicate the measured rotational angle.

the modelled and actual phase maps, indicating regions where the two maps misalign by fractions of π while also demonstrating regions where the model precisely determines the phase of the real data. The approach described in the previous section may also be feasibly applied to three-dimensional representations of the cortex, with its complex geometry and cortical folds. By ensuring the phase singularity configuration satisfies topological constraints, [62] this model may more accurately capture neural dynamics along the three-dimensional cortex.

Despite its predictive capabilities, this model is not without its limitations. It relies heavily on the accurate measurement of spiral wave rotational angle, which can significantly restrict the model's accuracy due to imprecision in this measurement. Additionally, this model relies on accurately and consistently identifying spiral wave centres from the real data. Misrepresentation of these spiral centres, such as duplicate identification of phase singularities demonstrated in the model of right foot movement in figure 4.2**(b)**, can disrupt the model's ability to measure the spiral wavefronts from the real data accurately and significantly inhibit the optimisation of the radial influence of the singularities.

The optimisation explored in this thesis is the beginning of what may be possible for this model. Simple gradient descent while effective, can be substantially improved upon by more advanced optimisation schemes. This is because simple gradient descent assumes uniform smoothness and struggles with non-smooth, non-uniform, or heavy-tailed landscapes, particularly relevant in complex functions such as this model.[65] A Genetic Algorithm, for instance, may leverage variance reduction and adaptability to local structures, enhancing the ability to find global minima for the convex cost landscape of this model. Furthermore, incorporating stochastic elements through a stochastic gradient descent would help navigate complex terrains more

effectively, thus achieving more reliable global optima than currently possible. With no standard methods of determining the radial influence of spiral waves in the brain, this model may serve as an innovative approach to addressing this vital issue by improving the optimisation schemes.

There are also some features of the complex spatiotemporal brain activity that this model does not currently capture specifically the dynamics of sink and source patterns. Although less frequent than spiral waves, these features may make regular appearances throughout the wave vector fields underlying the phase maps. These features sit outside the domain of this study; however, the interactions between spiral waves and source and sink points are a substantial area of future research.

As we conclude this chapter on the mathematical modelling of spiral wave interactions, it is essential to consider the transformative potential of our research. How the model can be further refined to integrate the functional significance and unique characteristics of spiral waves, as detailed in the previous chapters. Our current model provides a sophisticated framework for understanding spiral wave dynamics, yet it predominantly focuses on the geometric and topological aspects of these phenomena. To enhance the utility and applicability of the model, future revisions could incorporate findings from Chapter 3 which elucidate the task-specific roles and information-encoding capabilities of spiral waves during motor tasks. By integrating these functional aspects, the model could be expanded to simulate the physical presence of spiral waves and their dynamic roles in neural processing and cognition. This approach would allow us to explore how changes in spiral wave properties could affect neural function, offering deeper insights into the causal relationships between complex wave patterns and cognitive processes. Such enhancements would not only improve the descriptive power of the model but also its predictive capabilities, potentially opening new avenues for understanding and manipulating brain activity in both health and disease.

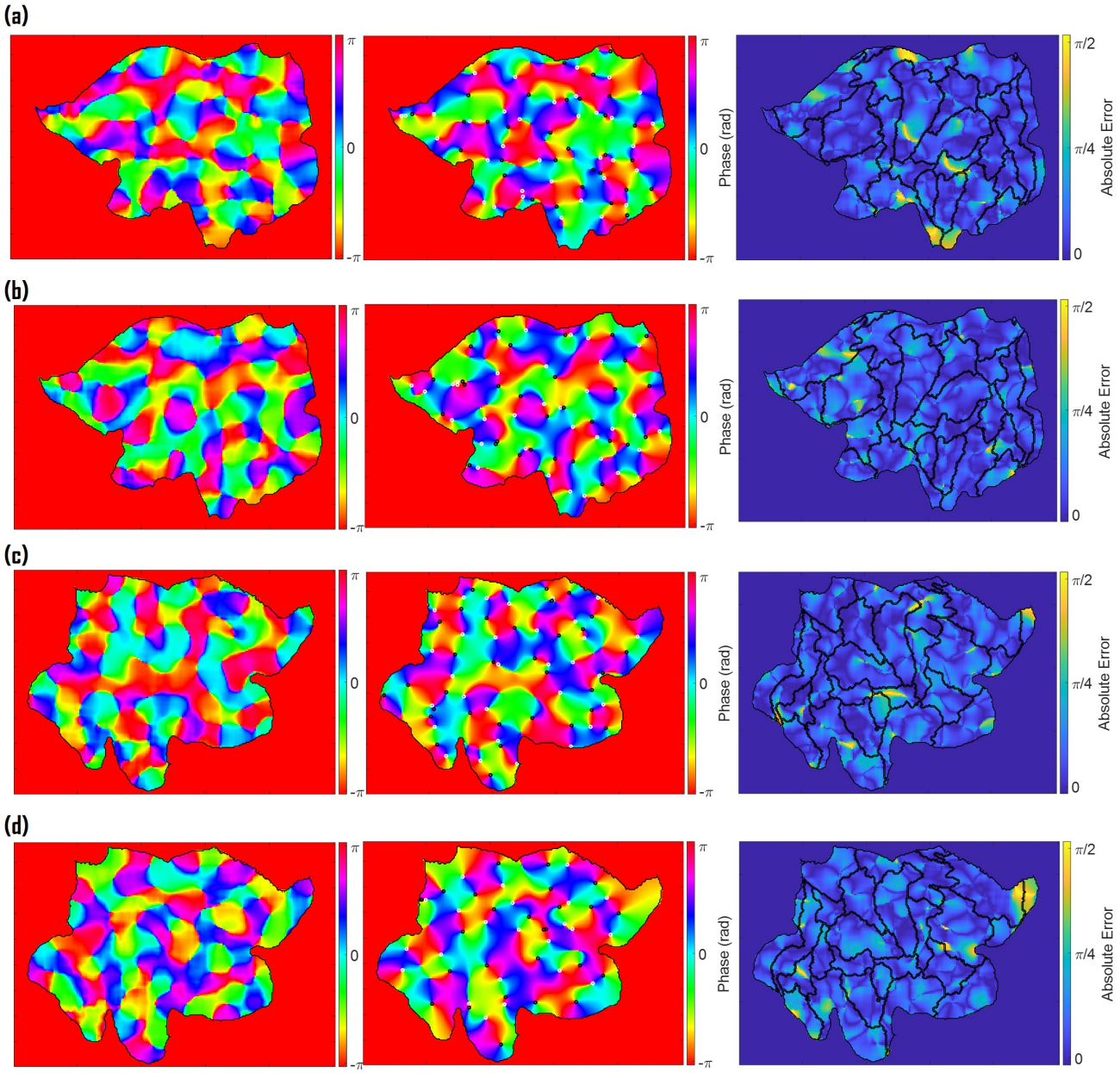

Figure 4.2: Comparing true & modelled phase maps For each row we show the actual phase data captured for an individual during a specific motor task, the recreation of this state by the model including the location of spiral wave centres and their charge (White: +1, Black: -1) and the absolute modular difference between the two maps. **(a)** Right Hand task, correlation coefficient: $r = 0.73$ **(b)** Right Foot task, correlation coefficient: $r = 0.76$**(c)** Left Hand task, correlation coefficient: $r = 0.76$ **(d)** Left Foot task, correlation coefficient: $r = 0.75$

Chapter 5

Discussion and Conclusion

This study investigates the hypothesis that spiral waves in the motor cortex are crucial for coordinating the neural activity underpinning motor functions. The analysis of spiral wave dynamics and their correlation with motor tasks reveals significant evidence supporting this hypothesis. The adaptability and variability of spiral wave responses to motor tasks demonstrate the need for further investigation into their adaptation across trials, indicating their role in the brain's learning processes and specific task demands. The persistence of these waves beyond initiating tasks highlights their ongoing influence on subsequent neural states, demonstrating their broader role in neural coordination. The findings on the lifetimes and propagation speeds of spiral waves provide further insights. The average lifetime of spiral waves aligns with half the duration of motor tasks, suggesting their influence during crucial phases of execution. Their propagation across cortical regions supports the hypothesis that spiral waves facilitate spatial distribution of neural coordination, linking cortical areas involved in motor control. Task-specific spiral wave density maps reveal distinct cortical locations for anti-clockwise and clockwise rotations depending on the motor task. This specificity provides evidence that spiral waves reflect the functional structures for different motor tasks. Less distinct patterns for foot movements suggest this structure is less visible, showing how spiral wave dynamics are specific to motor control needs. The classification of spiral centre locations based on task-specific patterns, with accuracies above chance levels, confirms that spiral waves are systematically related to task execution. Finally, the mathematical modelling through accurate recreation of phase singularities and their dynamics provides compelling validation for the dominance that these rotational structures may assert on the brain's overall structure. The model captures the complex interactions observed between spiral waves and replicates over half the variance in empirical data of motor functions, supporting a solid theoretical foundation for the role of spiral waves in motor cortex coordination.

The coordination of flow into specific activity hubs discussed in Chapter 3 links closely to the findings of functional networks discussed in the literature. The propagation of these waves demonstrates a transitionary quality that may underlie the localised activations and deactivations among functional regions in the pre-motor and motor cortex.[41] The plasticity of the motor cortex observed through functional connectivity also aligns with the variability of spiral wave response we observe in

Chapter 2. Specifically, studies have observed a significant reduction in functional connectivity between the pre-motor area and other motor-related regions following motor imagery training, suggesting a reorganisation of motor networks that we believe may be facilitated through the structural changes in the locations of spiral waves.[66] The appearance of these spiral waves within the motor cortex may also be used to explain the observations of dimensionality reduction techniques in mesoscale neural recordings.[10] The low-dimensional rotational manifolds distilled from projections of neural populations during motor tasks resemble cortical activity coordinated by the spiral waves we observe, pointing to a correlation between the two. By measuring spiral wave statistics, we draw parallels between the dynamics observed in the brain and complex systems, such as the biochemical wave patterns of living cells.[52] Through the model we create with minimal assumptions related to specific brain functions, phase singularity distributions are demonstrated as crucial encoders of their surrounding structure, mimicking their use in optics as singular skeletons to describe optical fields.[58][59]

The resolution limitations of fMRI data indeed constrain the findings detailed here. Future research could explore higher-resolution imaging techniques or real-time neural activity monitoring to capture these dynamics more accurately. The lack of behavioural data associated with these recordings is also a significant drawback in studying spiral wave dynamics. Measurements of reaction times and EMG recordings of muscle activity would allow for a more comprehensive understanding of the temporal relationship between neural activity and physical response. The higher precision guaranteed by optimising the model through more sophisticated techniques is crucial for further validating the structural encoding of phase singularities in brain states. Future work may also incorporate a three-dimensional representation of this model through the appropriate topological considerations.

Furthermore, given the detailed understanding of spiral wave dynamics in other fields, one could envisage developing reaction-diffusion equations that may evolve the initial conditions defined through our method, effectively predicting future brain states from the dynamic interactions of spiral waves. This would be revolutionary for computational models that simulate motor cortex activity. Applications of this predictive capability would be essential for developing targeted therapeutic interventions for neurological disorders such as epilepsy and Parkinson's disease. By leveraging the ability to foresee and manipulate brain activity patterns, medical professionals could significantly enhance treatment precision and outcomes. Moreover, such advancements also open new pathways for rehabilitating motor function impairments, providing a foundation for the next generation of brain-computer interfaces that adapt in real-time to the user's neural signals, thereby improving the efficacy and responsiveness of assistive technologies.

Overall, the results amassed from this study affirmatively answer the thesis hypothesis, demonstrating that spiral waves are fundamental to the coordination of neural activities in the motor cortex. Through their dynamic properties these waves significantly influence motor behaviour, confirming their pivotal role in both the mechanistic and functional aspects of brain dynamics during motor tasks. This work contributes a new perspective to computational neuroscience and aligns with the emerging trends in interdisciplinary research, combining complex systems theory and neuroscience to tackle fundamental questions about brain function.

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Appendix A

Code availability

The code for generating the mathematical model, along with code for producing the results in this thesis, are publicly available without restrictions on my fork of the Brain Vortex toolbox on Github at: https://github.com/AdamKarvon/BrainVortexToolbox